MILITARY SPECIFICATION

OSCILLATOR SUBASSEMBLY

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of an Oscillator Subassembly, Naval Ordnance Systems Command Drawings 156371, which is used in the enabler assembly of an acoustic panel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal.

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-E-1	Electron Tube and Crystal Rectifiers
MIL-C-25	Capacitors, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metallic case)
MIL-P-116	Preservation, Methods of
MIL-R-10509	Resistors, Fixed Film (High Stability) General Speci ticatians for
MIL-S-12883	Sockets, for Plug-in Electronic Components, and Accessories, General Specifications for

FSC 1355

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MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-200	Electron Tubes and Semiconductor Devices, Diodes
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts

DRAWINGS

Naval Ordnance Systems Command

DWG 1563731	Oscillator Subassembly

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect OR date of invitation for bide or requests for proposal shall apply.

National Bureau of Standards Publications

Handbook H28	Screw Thread Standards for Federal. Services

(Appications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

United States of America Standards Institute

USAS Y32.2 1967	Graphic Symbols for Electrical and Electronics Diagrams

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(Applications for copies should be addressed to DOD Single Stock Point, U.S. Naval Supply Depot, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Ascription. The oscillator subassembly is a sealed plug-in electronic circuit encased in a metal container. This subassembly contains a center-tapped inductor, two capacitors, and two resistors which are connected to form a variable frequency series-fed Hartley oscillator tuned circuit.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the oscillator shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing oscillators that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the oscillator shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563731.

3.2.4 Surface treatment, construction, and dimensions. Surface treatment, construction, and dimensions for the oscillator shall be in accordance with the applicative drawings, specifications, and publications listed in NAVORD Dwg 1563731.

3.2.5 Mositure, fungus, and corrosion. The oscillator shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in NAVORD Dwg 1563731.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in NAVORD Dwg 1563731.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

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3.2.8 Conflicing requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Symbols. Interpret schematic diagram symbols in accordance with USAS Y32.2 1967.

3.4 Performance requirements and product characteristics. The oscillater shall meet the following performance requirements and product Characteristics:

3.4.1 DC resistance

3.4.1.1 The direct current (dc) resistance between pins 2 and 7 of the oscillator shall be 330,000 ohms, plus or minus 5 percent.

3.4.1.2 The dc resistance between the coil center tap and pin 1 shall be 1800 ohms, plus or minus one percent.

3.4.2 Frequency. The frequency of the oscillator shall be 54,040 plus or minus 50 Hertz (Hz) with pin 6 ungrounded, and 51, 540 plus or minus 50 Hz with pin 6 gounded.

3.4.3 Voltage. The output voltage fromm the oscillator with pin 6 ungrounded shall be 2.75 plus or minus 0.2 volts root-mean- square (vrms). With pin 6 grounded, the voltage shall be 2. 50 plus or minus 0.2 vrms.

3.4.4 Waveform. The oscillator output voltage shall be sinusoidal and the distortion factor shall not exceed two percent. The distortion factor, D, is given by the expression

$$D(\text{percent}) = 100\sqrt{\frac{\text{sum of the squares of amplitudes of harmonics}}{\text{square of amplitude of fundamental}}}$$

3.4.5 Dielectric strength. There shall be no evidence of breakdown when 200 vrms, 60 Hz is applied between insulated terminals and case for l period of one minute,

3.4.6 Insulation resistance. The imsulation resistance shall be 10,000 megohms minimm be tween insulated terminals and case when measured at l test potential of 100 volts direct current (vdc). The insulation resistance measurement shall be made immediately after the dielectric strength test potential is removed.

3.4.7 Operating life. Life. The oscillator shall have an operating life of at least 800 cycles when tested as specifiedin 4.5.5.

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3.5 Environmental requirements. The oscillator subassembly shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.5.1 Vibration. The oscillator subassembly shalll meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.5.1.2 Vibration, operating. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.5.1.3 Vibration, transportation. The oscillator subassembly shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.4.

3.5.2 Shock, transportation. The ascillator subassembly shall shi[n]- no damage and shall meet all the -performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.5.3 Temperature and humidity

3.5.3.1 High temperature and humidity, nonoperating. The oscillator subassembly shall show no damage and shall reset all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.5.3.2 High temperature, operating The oscillator subassembly shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3. 5.3.3 Low temperature, nonoperating The oscillator subassembly shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

3.5.3.4 Low temperature operating The oscillator subassembly shall be capable of meeting all the performance requirements of this specification while undergoing the test specified In 4.5.3.4.

3.5.4 Immersion. The oscillator shal 1 show no ev idence of leakage during the immersion test specified in 4.5.4. Following the test, the oscillator shall meet the performance requirements and product characteristics of this specification.

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3.6 Cleanup. Prior to and after final assembly all parts, components, and the assambly shall be thoroughly Cleaned of loose, spattered, or @L-cess solder, metal chips and other foreign rotter. Burrs and sharp edges as well as excessive rosin flash shall be removed.

3.7 Workmanship. The oscillator, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of thi s specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of pints and as semblies, plating, painting, riveting, machine screw as semblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Govenment's notice of approval, shall be determinative of the re-quirements of the contract relative to workmanship insofar as not specif icallly covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the perform-ance of all inspection requirements as specified herein. Except as other-wise spec ified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection re-quirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspecti ons are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the oscillator shall be classified as follows:

(a) Preproduction inspection (paragraph 405)

(b) Quality conformancee inspection (paragraph 4.4)

(C) Periodic production inspection inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five asillators, manu-factured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be basedf on no defects in the sample. Further production of the oscillators by the supplier, prior to the approval of the preproduction sample, shall be at the supplier 's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate “inspection lot” and

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may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production. samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five oscillators will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The oscillator: shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each o [illegible] delivered under contract shall be subjected to the following t
Failure to satisfactorily pass any test specif
cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the oscillator shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature. Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18.33 degrees Centigrade (C) to 35 degrees C).

4.4.1.2 Altitude. Normal ground.

4.4.1.3 Vibration None.

4.4.1.4 Humidity. Room ambient to 95 percent relative, maximum.

4.4.2 Test and inspection equipment facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performancee of the required acceptance tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance inspection set forth in this specification:

4.4.3,1 Power supply capable of supplying the test circuit shown in Figure 1 with electrical power conforming to the following requirements:

4.4.3.1.1 Filament power: The electrical power used to operate the

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6C4WA filament in the test circuit shall meet the following requirements:

4.4.3 .1.1.1 No-load voltage: With no-load the voltage shall not exceed minus 7.5 volts direct current (vdc).

4.4.3.1.1.2 Current: Filament current shall be 0.15 ampere.

4.4.3.1.1.3 Voltage: Under a load current of 0.15 ampere the voltage shall be 6.3 plus or minus 0.1 vdc.

4.4.3.1.1.4 Internal impedance: The impedance which the power source presents to the filament at a load current of 0.15 ampere shall not exceed 3000 ohms at a frequency of 53 kiloHertz (KHz)

4.4.3.1.1.5 Alternating current (at) voltage components: Under a load current of 0.15 ampere, the ac volage components shall not exceed the following:

(a) 30 Hz to 150 KHz; 0.09 volts root-mean-square (rms)

(b) 50 KHz to 56 KHz; 40 microvolt rms

4.4.3.1.2 Plate power: The electrical power used to operate the 6CWA pate in the oscillator test circuit shall meet the following requirement 6:

4.4.3.1.2.1 no-load voltage: With no load the voltage shall not exceed 3.65 vdc.

4.4.3.1.2.2 Current: Plate current shall be 4 millampers (ma).

4.4.3.1.2.3 Voltage: Under a load current of 4 ma the voltage shall be 150 plus or minus 1 vdc.

4.4.3.1.2.4 Internal impedance: The impedance which the power source presents to the test circuit at 8 load current of 4 ma shall not exceed 5 ohms Et 53 KHz

4.4.3.1.2.5 AC voltage components: Under a load current of 4 ma, the ac voltage components shall not exceed the following:

(a) 30 Hz to 150 KHz; 0.2 volts rms.

(b) 50 KHz to 56 KHz; 20 microvolts rms.

4.4.3.2 Electronic counter: Hewlett-Packard Model 523C, or equivalent.

4.4.3.3 Electronic voltmeter with probe having a 7.5 picofarad or less input capacitance and an output jack for connecting it to the electronic

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counter: Ballantine Model 314 or 314A, with probe, is satisfactory.

4.4.3.4 Insulation tester: Radio Frequency Laboratories Model 184C, or equivalent.

$.4.3.5 Wave analyzer: Sierra Model 121, or equivalent.

4.4.3.6 Oscilloscope: Tektronix Model 531 with 53/54C plug-in unit, or equivalent.

4.4.3.7 Dielectric tester: MOD NMD/Y.

4.4.3.8 power source: 115 wins, 60 HZ.

4.4.3.9 capacitor: O.1 microfarad, plus or minus 5 percent, 200 vdc; MIL-C-25 designation CP09AIKC104K.

4.4.3.10 Electron tube: 6C4WA, MIL- STD-200 designation; JAN-6C4WA. This tube shall have characteristics that are in the center of the limits plus or minus 5 percent, when tested as specified in MIL-E-1/857.

4.4.3.11 Resistor: 33,000 ohms plus or minus 1/2 of 1 percent, 0.5 watt; MIL-R-105O9.

4.4.3.12 Resistor: 0.33 megohms, plus or minus 1/2 of 1 percent, 0.5 watt; MIL-R-10509.

4.4.3.13 Resistor: 3.0 megohms, plus or minus 1/2 of 1 percent, 0.5 watt; MIL-R-10509.

4.4.3.14 Socket , tube : 7 pin miniature, with shield; MIL-S-12883 designation TS102P01.

4.4.3.15 Socket, tube: 7 pin miniature, without shield; MIL-S-12883 designation TS102P02.

4.4.3.16 Shield, miniature: MIL-S-12883 designation TS102U02.

4.4.3.17 Audio oscillator. Hewlett-Packard Model 200 CD, or equivalent.

4.4.3.18 Test circuit wiring capacities: With power removed, the 6C4WA in its socket with shield in place, oscillator subassembly removed, and inters disconnected, the following capacities to ground at the oscillator subassembly socket shall exist:

(a) pin 1 to ground; 9.6 plus or minus 0.5 micromicrofarads (μμf)

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(b) pin 2 to ground; 7.5 plus or minus 0.5 μμf

(c) pin 6 to ground; 2.0 plus or minus 0.5 μμf

(d) pin 7 to ground; 3.0 plus or minus 0.5 μμf

4.4.3.19 Resistance bridge: Leads-Northrup, Type S, Test Set No. 5300, or equivalent.

4.4.3.20 Connectors, test leads, and small tools as required.

4.4.3.21 Impedance bridge: John Fluke Model 710-A, or equivalent.

4.4.3.22 Power amplifier: McIntosh Model 75, or equivalent.

4.4.4 Test procedure. The following tests constitute the quality conformance tests for the oscillator:

4.4.4.1 Unless otherwise specified, all performance tests shall be conducted with the test equipment given in 4.4.3 and shall accurately follow the test procedure of 4.4.

4.4.4.2 Preparation for tests. Connect the power supply (4.4.3.1)) electronic counter 4.4.3.2) voltmeter (4.4.3.3), insulation tester (4.4.3.4), wave analyzer (4.4.3.5), oscilloscope (4.4.3.6) and dielectric tester (4.4.3.7 to the power source (4.4.3.8). Set the power switches of these instruments to ON and allow them to warm up for the time necessary to stabilize.

4.4.4.3 Dielectric strength. To measure the dielectric strength as specified in 3.4.5:

4.4.4.3.1 Connect the dielectric tester to pin 1 and the case of the oscillator and apply 200 vrms at 60 Hz for one minute. There shall be no evidence of breakdown.

4.4.4.3.2 Repeat 4.4.4.3.1 for each remaining pin (pins 2, 3, 6, and 7). of the! oscillator.

4.4.4.4 Insulation' resistance. To measure the insulation resistance as specified in 3.4.6:

4.4.4.4.1 Disconnect the diaelectric tester Leads and connect the insulation tester to pin 1 and the case of the oscillator. Apply 100 vdc test potential for two minutes and then measure the insulation resistance at this teat potential. The Insulation resistance shall be not less than 10,000 megohms.

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4.4.4.4.2 Repeat 4.4.4.4.1 for each remaining pin (pins 2, 3, 6, and 7) of the oscillator.

4.4.4.5 DC resistance as set forth in 3.4.1.2:

4.4.4.5.1 Calibrate the resistance bridge to measure resistance in accordance with the instruction provided with that instrument.

4.4.4.5.2 Measure the resistances from pins 1 to 2, 1 to 3, and 2 to 3 of the oscillator. Compute the value of the resistance between the coil center tap and pin 1 by the expression

$$R = \frac{(R_{1 \text{ to } 2} + R_{1 \text{ to } 3}) - R_{2 \text{ to } 3}}{2}$$

This resistance shall be 1800 ohms, plus or minus 1 percent.

4.4.k.6 DC resistance as set forth in 3.4.1.3.:

4.4.4.6.1 Measure the resistance from pin 2 to pin 7 of the oscillator. This resistance shall be 330,000 ohms, plus or minus 5 percent.

4.4.4.6.2 Remove the resistance bridge test leads.

4.4.4.7 Frequency. To measure the frequency as set forth in 3.4.2:

4.4.4.7.1 Remove power from the power suppply. Connect the oscillator test circuit as shown in Figure 1. Place the 6C4WA tube properly in its socket, and remove the oscillator from its socket. Check the test circuit wiring capacitances between pins 1, 2, 6, and 7 of the oscillator socket and ground to ensure that they are in accordance with 4.4.3.18.

4.4.4.7.2 Replace the oscillator in its socket and apply power to the power supply and test circuit, allowing enough warmup time for the equipment to stabilize.

4.4.4.7.3 Check the calibration Of the electronic counter in accordante with the instructions provided with that instrument. when the counter has been satisfactorily checked, set the controls to measure frequency.

4.4.4.7.4 Connect the counter imput as shown in Figure 1. Read and record the frequenc y indicated on the counter. This frequency shall be 54,040 plus or minus 50 Hz.

4.4.4. 7. 5 Attach the ground clip to pin 6 of the oscillator socket. Read and record the frequency y Indicated on the counter. This frequency shall be 51,540 plus or minus 50 Hz.

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4.k.4.8 Voltage. To measure the voltage as set forth in 3.4.3:

4.4.4.8.1 Remove power from the power supply. Connect the oscillator test circuit as shown in Figure 2. Place the 6C4WA tube properly in its socket and remove the oscillator from its socket.

4.4.4.8.2 Adjust the audio oscillator (4.4.3.17) to apply 30 vrms at a frequency of 54040 + 50 Hz to pin 7 of the oscillator socket. Read and record the voltmeter #1 indication, which. should be approximately 2.75 volts.

4.4.4.8.3 Adjust the audio oscillator (4.4.3.17) to apply 30 vrms at a frequency of 515 40 ± 50 Hz to Pin 7 of the oscillator socket. Read and record the voltmeter #1 indication, which should be approximately 2.50 volt s l

4.4.4.8.4 Disconnect the input signal. to pin 7 of the oscillator socket and replace the oscillator in its socket to connect the oscillator test circuit as shown in Figure 1. Apply power to the power supply and test circuit, allowing enough warmup time for the equipment to stabilize.

4.4.4.8.5 Connect the voltmeter (4.4.3.3) as shown in Figure 1. Read and record the voltmeter indication. This voltage shall be within plus or minus 0.2 vrms of the voltage measure in 4.4.4.8.3.

4.4.4.8.6 Remove the ground clip from pin 6 of the oscillator socket. Read and record the voltage indicated on the voltmeter. This voltage shall be within plus or minus 0.2 vrms of the voltage measured in 4.4.4.8.2.

4.4.4.8.7 Remove the voltmeter test Leads.

4.4.4.9 Waveform To check the wave form as set forth in 3.4.4:

4.4.4.9.1 Connect the oscilloscope vertical input to pin 7 of the oscillator socket. Connect the vertical input shield to pin 3 of the same socket. Adjust the oscilloscope controls for a good presentation. Observe the waveform. The waveform shall be sinusoidal and shall not contain noise or spikes (transients).

4.4.4.9.2 Remove the oscilloscope vertical input leads from pin 7 and connect the wave analyser input to pin 7 and the shield or low side to pin 3 of the oscillator socket. In accordance with the instructions provided with the wave analyzer, measure and record the amplitude of the fundamental and the amplitudes Or as many harmonics as possible.

4.4.4.9.3 Using the values recorded in 4.4.4.9.2 and the expression given in 3.4.4, compute the distortion factor. The distortion factor, D, shall not exceed two percent.

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4.4.4.9.4 Remove the wave analyzer test leads and secure the power to the test equipment and power supply.

4.4.4.10 Visual and dimensional examinations. Each oscillator shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4. 11 Packaging, packing) and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Reproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The oscillator subassembly, less pack or packaging, shall be mounted to a test fixture as shown in Figure 3 and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The oscillator subassembly shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

4.5.1.3 Vibration, operati ng. The oscillator subassembly shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.5.1.2.

4.5.1.4 Vibration, transportation. The oscillator subassembly shall be subjected to the "Vibration, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.3.

4.5.2 Shock, transportation. The oscillator subassembly shall be subjected to the "Shock, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.

4.5*3 Temperature and humidity

4.5.3.1 High temperature and humidity, nonopera ting. The oscillator subassembly shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requiements of 30503.10

4. 5.3.2 High temperature, operating. The oscillator subassembly shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.2.

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4.5.3.3 Low temperature, nonoperating. The oscillator subassembly shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.3.

4.5.3.4 Low temperature operating. The oscillator subassembly shall be subjected to the Low temperature; operating" test of MIL-E-82590 and, while at the test temperatureshall meet the requirements of 3.5.3.4

4.5.4 Immersion. The oscillator shall show no evidence of leakage while being subjected to the immersion test specified in MIL-STD-202, Method 104, Condition A, and thereafter shall meet the performance requirements and product characteristics of this specification.

4.5.5 Operating life. The operating life of the oscillator subassembly shall be tested as follows.

4.5.5.1 connect the oscillator subassembly as shown in Figure 1. Apply filament power to the vacuum tbe and allow it to warm up for at least 30 seconds. Apply 150 plus or minus 1 vdc as shown in Figure 1, and allow the circuit to operate for 30 plus or minus 1 minutes. Then remove the 150 vdc and allow the circuit to rest for at least one hour . This shall be considered one cycle of operation.

4.5.5.2 Repeat 4.5.5.1 until 800 cycles of operation have been completed, after which the oscillator subassembly shall meet the requirements of 3.4.1 through 3.4.4.

4.5.5.3 The vacuum tube and test circuit components maybe replaced as necessary to permit the continance of this test.

4.6 Waiver of tests. Tests 'specified in 4.5.1.4 and 4.5.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Fundcional tests. At the conclusion of the tests specified in 4.5.1.4 and 4.5.2 above, functional and other tests, as applicable, shall be conducted on the oscillator subassembly to determine freedom from operational malfunction and compliance with all performance requirement and product characteristics of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The oscillator shall be packaged in accordance with Method IIB of MIl-P-116.

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5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging The oscillator shall be packaged in accordance with Method III of MIL-)-116.

5.2 Packing. Patti.zuz shall be Level A or Level Ç, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The oscillators, packaged as described in 5.1.1.1, shall be packed in a wooden lox in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 95.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The oscillator, packaaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing, or other adequate shock-absorbing devices to ensure that the oscillator will meet all the requirements of Section 3 after being tested in accordance with paragraph 4.5.1.4 (Vibration) and 4.5.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. in addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The oscillator is Intended for use in Torpedoes Mk 37 Mods 0, 2 and 3.

Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level Of packings, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other then that specified in 3.2.1.

(g) That additional sanples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

(h) Guarantee clause (see 6.3).

6.2.1 The attention of the contracting officer is invited to the options in 4.6.

6.3 Guarantee. The contract or order shall include a requirement for a guarantee stating that any oscillator which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

Preparing Activity
Navy - OS
Project No. 1355-N 217

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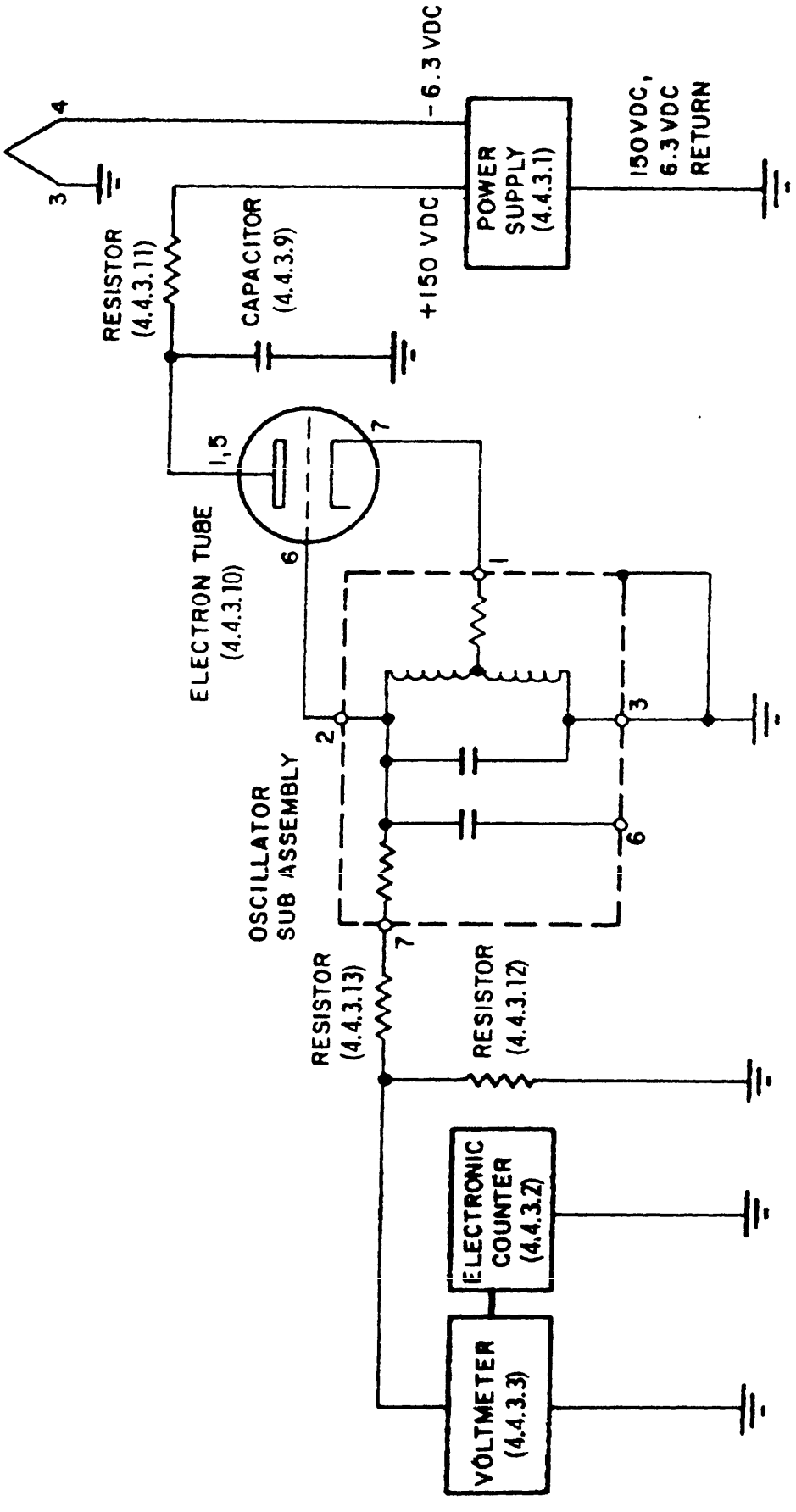


Figure 1. Oscillator Subassembly Test Circuit.

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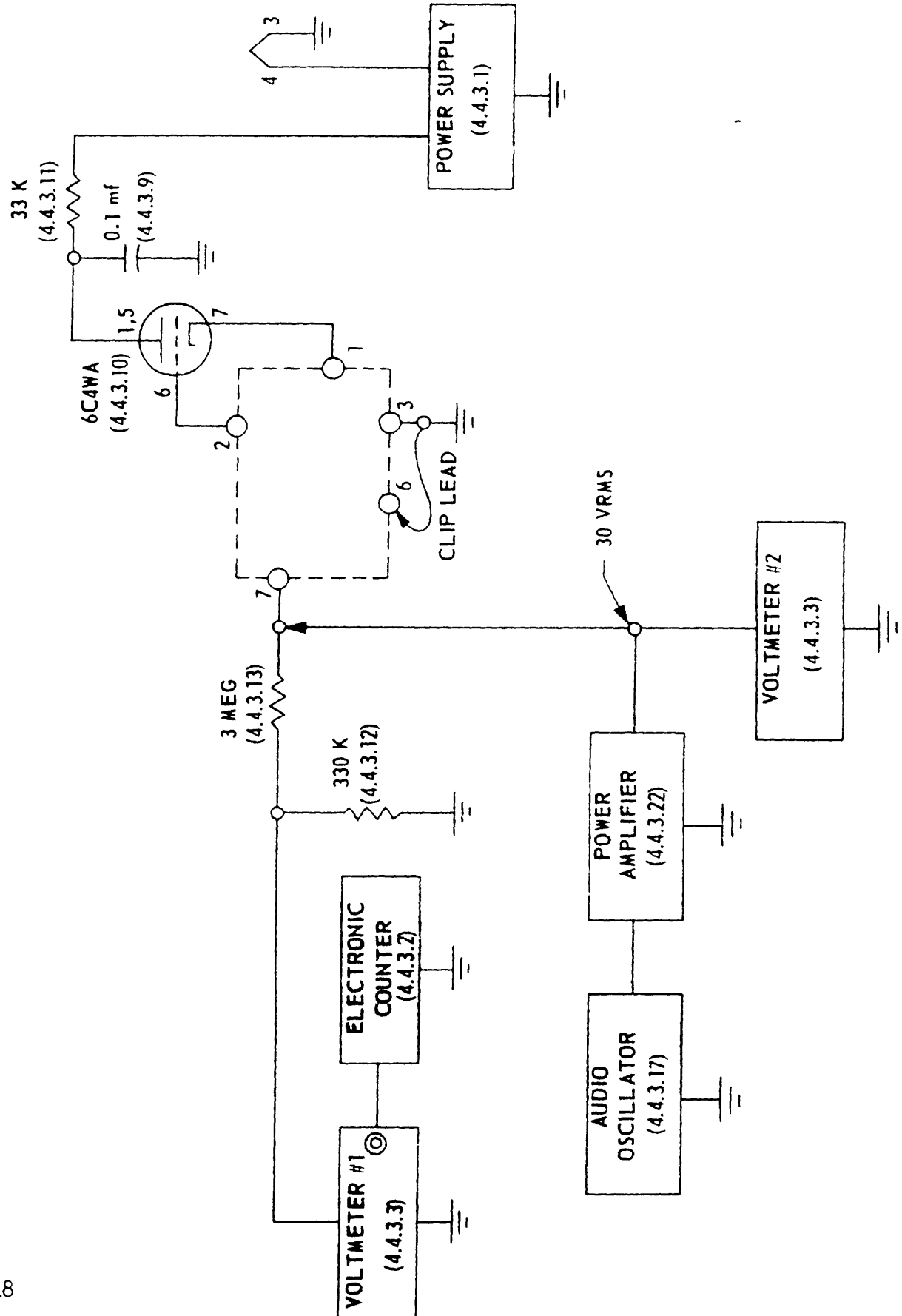
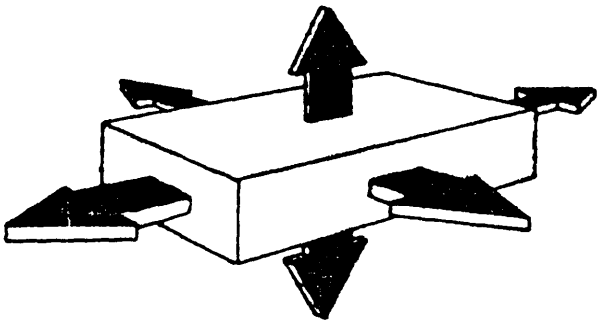


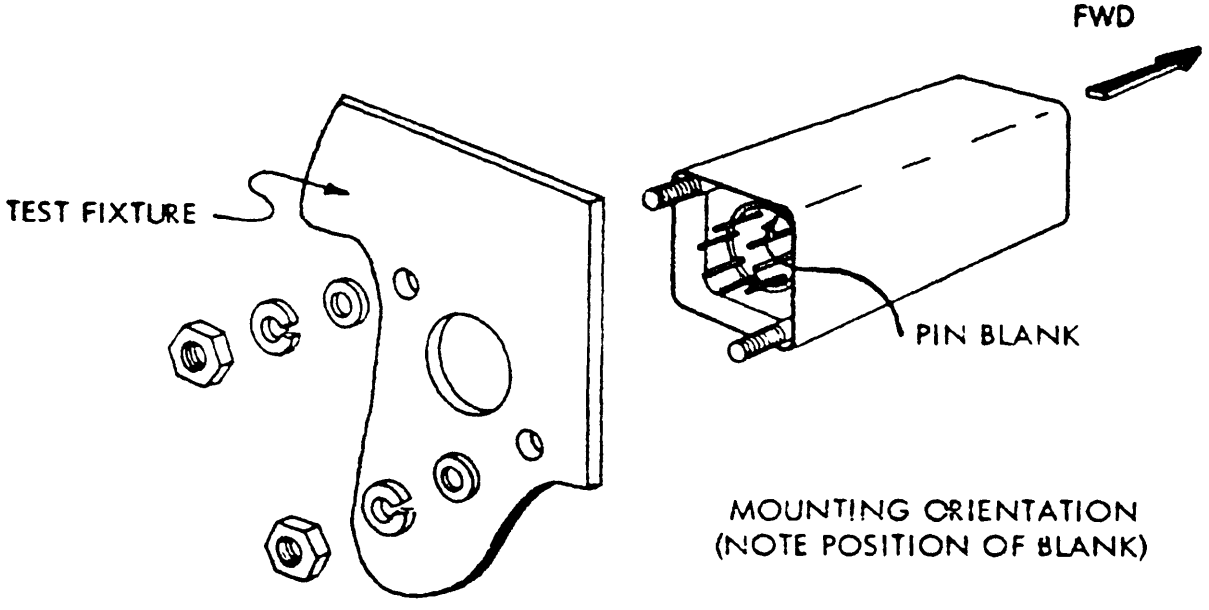
Figure 2. Test Circuit Number Two.

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DIRECTIONS OF VIBRATION INPUTS





MOUNTING ORIENTATION
(NOTE POSITION OF BLANK)

NOTES:

1. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS.

2. FOR VIBRATION, THE TEST ITEM SHALL BE SECURED TO THE TEST FIXTURE AS SHOWN.

3. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

Figure 3. Oscillator, NAVORD Dwg NO. 1563731, Mounting Requirements.

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